

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Design advanced RAG pipelines (multi-stage retrieval, hybrid search, re-ranking) combining GraphRAG with vector-based retrieval across multiple LLMs (GPT, Claude, Llama, Gemini) for grounded, multi-hop enterprise queries.
- Own multi-step agentic workflows (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK) with MCP-compatible tool integrations and stateful agent memory via Redis checkpointing at 99%+ uptime.
- Built a deployment-gated AI evaluation framework covering model quality, latency, and cost, enforcing guardrails and observability, blocking 15% of failed releases before production.
- Build scalable platform APIs (Python, FastAPI, TypeScript/React) on AWS and Azure, partnering with product and business stakeholders to ship production AI features end-to-end.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Ran hands-on model training/fine-tuning (PyTorch) on multi-GPU clusters with compression and quantization for efficient production inference.
- Built custom evaluators for RAG/agent workflows tracking drift, latency, and regression across model versions, catching failures before they reached production.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with MCP-compatible tool-use compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP server for enterprise data queries.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++

Agentic AI & LLMs: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, GPT, Claude, Llama, Gemini, MCP, Tool Calling

Retrieval & Evaluation: RAG, GraphRAG, Multi-Stage Retrieval, Re-Ranking, Model/Agent Evaluation, Guardrails, Observability, Cost Optimization

Cloud & Platform: AWS, Azure, GCP, FastAPI, Docker, Kubernetes, PostgreSQL, Redis, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer